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End-To-End Deep Learning System for Stuck Pipe Prevention in Drilling Operations

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Abstract

This paper introduces an end-to-end deep learning system designed to proactively detect and classify anomalies enabling stuck pipe incident detection well in advance of the actual stuck pipe incidents in drilling operations. The system operates directly on raw sensor data without preprocessing or reliance on predefined rig states.

The proposed system integrates specialized models targeting the tripping and drilling phases separately. It continuously monitors sensor data to detect anomalies such as overpull, overslack, restrictions, stick and slip, pipe stalling, and pressure abnormalities. Anomalies are classified by severity (low to high), supported by a dynamic stuck pipe risk indicator that escalates with increasing severity. The deep learning architecture, named DrillNet, is specifically optimized for drilling applications.

Evaluated on a large dataset of 200 drilled sections—significantly larger than datasets used in typical methods—the system achieves an F1 score above 70%. Compared to conventional techniques, it delivers improved accuracy, quicker responsiveness, and stronger generalization. Real-world implementation has confirmed its effectiveness in early anomaly detection, significantly reducing downtime, improving operational safety, and cutting operational costs.

The novelty lies in using raw sensor data directly without preprocessing or predefined rig states, designing a deep learning architecture (DrillNet) and the extensive dataset for evaluation (200 drilled sections) substantially exceeding industry norms, enhancing the reliability and applicability of the solution.

Introduction

Background

Stuck pipe has been a major problem in drilling even when most wells were drilled vertically. The evolution of stuck pipe prevention has progressed through several technological phases. Early mechanical approaches focused on equipment modifications and drilling fluid optimization (Bradley, 1979; Mitchell, 2009). The introduction of real-time monitoring systems in the 1990s enabled continuous surveillance of drilling

parameters (Johancsik et al., 1984), while advanced torque-drag modeling provided improved predictive capabilities (Sheppard et al., 1987).

Stuck-pipe incidents remain among the most critical and costly challenges in petroleum drilling operations, often accounting for a substantial share of well costs and contributing significantly to non-productive time (NPT). These events can result in millions of dollars in additional expenses, and despite decades of technological progress, their unpredictable nature continues to hinder drilling efficiency and pose serious safety risks (Elmousalami & Elaskary, 2020).

Traditional approaches have relied on reactive measures and torque-drag models that provide limited real-time predictive capabilities (Salminen et al., 2017; Al-Mamoori et al., 2025). Recent machine learning research has shown promise, with artificial neural networks, support vector machines, and ensemble methods for stuck pipe prediction (Al-Baiyat & Heinze, 2012; Elahifar & Hosseini, 2022; Al-Mamoori et al., 2025). However, existing approaches face critical limitations: dependence on preprocessed data and predefined rig states, and lack of comprehensive real-time deployment capabilities.

Drilling analytics has also evolved more broadly across industry applications. Early efforts began with rule-based rig-state detection, which later transitioned to learned models that improved state classification from surface sensors, providing value for downstream KPIs (Benzine et al., 2024a). More recently, industrial-scale models have automated daily drilling report (DDR) codification and quality control (Benzine et al., 2024a). In parallel, real-time anomaly detection systems flag critical events such as overpull (Benzine et al., 2024b). Multimodal methods now align drilling time-series segments with report text for retrieval and supervision (Buiting et al., 2025). At the same time, practical deployment constraints often favor compact models, such as 1D CNNs, which perform well under latency limits in real-world settings (Benzine et al., 2025). Recent work has further explored the role of large foundation models for time series segmentation in industrial drilling settings (Khaouja et al., 2025).

Recent advances in machine learning have shown significant promise in drilling applications. Artificial neural networks have been successfully applied to drilling optimization and anomaly detection (Bilgesu et al., 1997; Elahifar & Hosseini, 2022). Support vector machines have demonstrated effectiveness in stuck pipe prediction (Noshi et al., 2018; Al-Baiyat & Heinze, 2012). Ensemble methods combining multiple algorithms have further improved prediction reliability (Sabah et al., 2019). Recent developments in AI-automated drilling systems have demonstrated the integration of machine learning and deep learning approaches for real-time operations center (RTOC) applications, showing promising results in automated anomaly detection during drilling operations (Benzine et al., 2024a).

Current methodologies require extensive feature engineering and domain-specific preprocessing, potentially losing subtle patterns in raw sensor data that may indicate impending stuck pipe events. Most studies treat stuck pipe prediction as binary classification without considering the progressive nature of anomaly development or providing granular risk assessment.

This paper introduces a novel end-to-end deep learning system for proactive stuck pipe prevention that operates directly on raw sensor data without preprocessing requirements. Our approach integrates specialized models for different drilling phases, provides continuous monitoring with comprehensive anomaly classification (overpull, overslack, restriction, stick and slip and pipe stalling), and includes a dynamic risk indicator with severity-based escalation.

Key contributions include:

1. DrillNet, a specialized deep learning architecture optimized for drilling applications;
2. A comprehensive anomaly classification with real-time risk assessment; and
3. A demonstrated real-world implementation with measurable improvements in operational safety and cost reduction.

The system achieves an F1 score above 70% while processing raw sensor data, representing a significant advancement in practical stuck pipe prevention capabilities.

Problem Definition

The stuck pipe prevention problem can be formally defined as a multi-stage time series classification and segmentation task operating on raw drilling sensor data, and it helps monitor the risk of stuck pipes by looking for preceding anomalies (cf. Table 1). Given a sequence of sensor measurements $X = \{x_1, x_2, \dots, x_t\}$ collected at time steps t , where each x_i represents a multi-dimensional vector of drilling parameters (hook load, torque, standpipe pressure, weight on bit, rotary speed, flow rates, etc.), the objective is to learn a mapping function f that can simultaneously predict:

- 1. **Binary Anomaly Detection:** $f_1(X) \rightarrow \{0, 1\}$, determining whether anomalous drilling conditions are present
- 2. **Multi-class Category Classification:** $f_2(X) \rightarrow \{\text{overpull, slack-off, restriction, pipe stalling, stick and slip and stuck pipe}\}$, identifying the specific type of anomaly when detected
- 3. **Severity Assessment:** $f_3(X) \rightarrow \{\text{low, medium, high}\}$, quantifying the severity level of the identified anomaly

Table 1—Categories of Anomalies with their corresponding Severities

Category	Severity
Over-pull	low
	medium
	high
Over-slack	low
	medium
	high
Restriction	medium
	high
Pipe stalling	medium
	high
Stick and slip	low
Stuck pipe	high

The challenge lies in developing a unified architecture that can process raw, unprocessed sensor data in real-time while maintaining high accuracy across all three prediction tasks. The system must be robust to the inherent noise and variability in drilling operations, capable of generalizing across different well profiles and drilling conditions, and responsive enough to provide early warnings before stuck pipe incidents occur.

Unlike traditional binary classification approaches that simply predict stuck/non-stuck outcomes, this formulation captures the progressive nature of stuck pipe development through hierarchical anomaly detection, enabling proactive intervention at multiple stages of the incident evolution.

Data Description

Our anomaly detection model ingests a comprehensive set of real-time surface parameters acquired via WITSML (Wellsite Information Transfer Standard Markup Language) at approximately 1Hz sampling frequency. This high-fidelity data stream enables a holistic representation of the drilling process, facilitating the identification of diverse anomaly types.

The key parameters encompass:

Table 2—Key Parameters Acquired via WITSML and their Descriptions

Data	Units	Description
Bit Depth	Feet (ft)	The current depth of the drill bit from the surface
Hole Depth	Feet(ft)	The total depth of the wellbore
Hook Load	Kilo-Pound Force (kip)	The weight supported by the hook, reflecting the weight of the drill string minus the weight on bit
Block Position	Feet(ft)	The height of the traveling block in the derrick
Stand Pipe Pressure	Pounds per square inch (psi)	The pressure of the drilling fluid as it enters the drill string
Flow Rate	US Gallon per minute (us gpm)	The volume of drilling fluid being pumped into the wellbore
Rotary Speed	Rotation per minute (RPM)	The rotational speed of the drill string
Torque	Kilo foot-pounds (kft-lbs)	The rotational force applied to the drill string

The Figure 2 below illustrates all these sensor data:

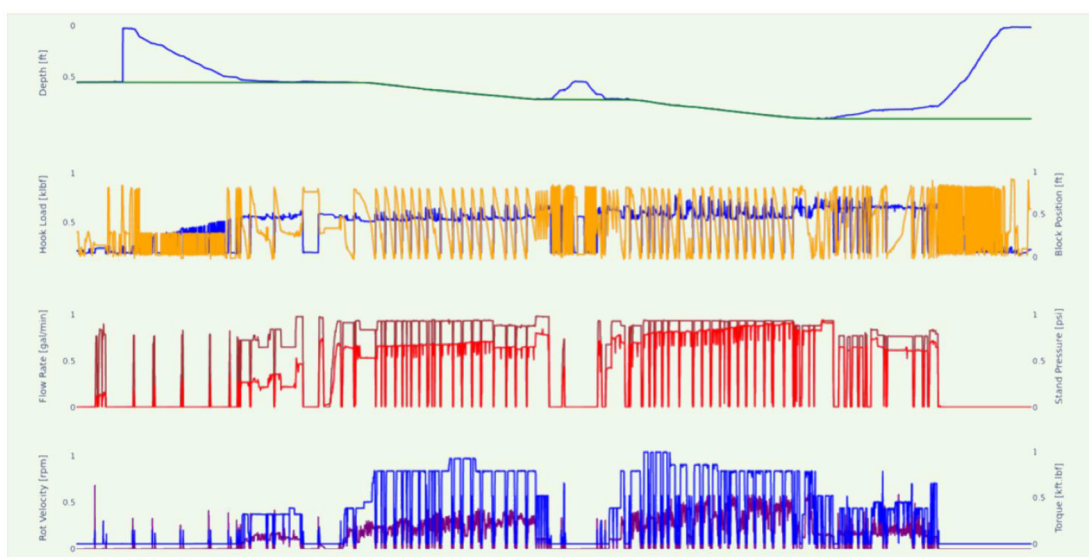


Figure 1—Example of drilling sensor data. First row: bit depth and hole depth. Second row: hook load and bloc position. Third row: Flow Rate and Stand Pressure. Last row: RPM and Torque

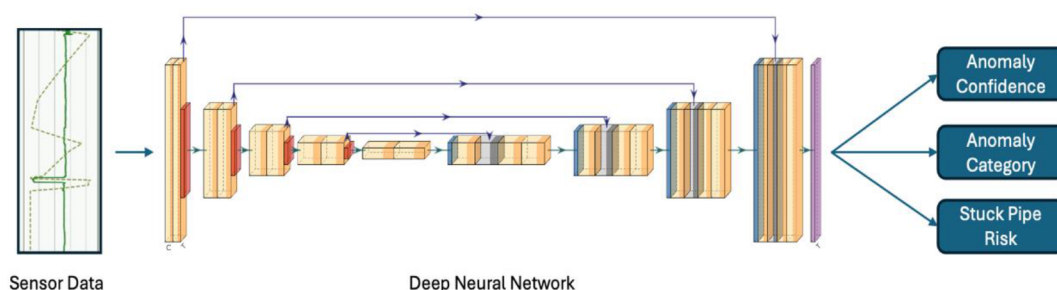


Figure 2—Multi head Deep learning Architecture

This comprehensive set of real-time surface data provides the foundation for our anomaly detection model to tackle the challenges outlined in the problem definition. By ingesting and processing this high-fidelity,

multi-dimensional data stream, our system can identify a wide range of drilling anomalies, including both well-established patterns and more subtle, complex deviations that have historically evaded detection by traditional methods.

Methodology

System Architecture Overview

Deep learning architectures have shown exceptional performance in time series analysis and anomaly detection. Convolutional neural networks excel at capturing temporal patterns in sequential data (LeCun et al., 2015), while encoder-decoder architectures have proven effective for sequence-to-sequence learning tasks (Sutskever et al., 2014). Recent work in industrial applications has demonstrated the effectiveness of fully convolutional networks for time series classification (Wang et al., 2017) and multi-head architectures for hierarchical prediction tasks (Vaswani et al., 2017).

The proposed end-to-end deep learning system for stuck pipe prevention is built around a multi-head architecture that processes raw sensor data to simultaneously predict anomaly confidence, classify anomaly categories, and assess stuck pipe risk (cf. Figure 2). Unlike traditional approaches that rely on preprocessed data and predefined rig states, our system operates directly on time-series sensor data, enabling real-time anomaly detection without manual feature engineering.

The system architecture comprises three main components:

1. a fully convolutional neural network (FCNN) backbone with encoder-decoder structure for feature extraction,
2. a multi-head output layer for simultaneous prediction of multiple targets, and
3. a dynamic risk assessment module that integrates predictions to generate actionable insights for drilling operators.

DrillNet Architecture

Encoder-Decoder Backbone. The core of our DrillNet architecture employs a fully convolutional neural network with an encoder-decoder structure, which has been previously proven as a strong benchmark for drilling-related time series analysis. The encoder component applies 1D convolutions and temporal pooling operations to capture multi-resolution patterns in the raw sensor data, effectively learning hierarchical representations at different temporal scales.

The encoder processes input sequences through multiple convolutional layers with progressively increasing receptive fields, allowing the network to capture both short-term fluctuations and long-term trends in drilling parameters. Temporal pooling operations reduce the dimensionality while preserving critical temporal information, enabling efficient processing of extended time series data.

The decoder component up-samples the encoded features through convolutions, restoring the original temporal resolution. Skip connections between corresponding encoder and decoder layers preserve fine-grained temporal details that might be lost during the encoding process, ensuring that the network maintains sensitivity to subtle anomalous patterns.

Multi-Head Output Structure. The DrillNet architecture incorporates a multi-head output design that enables hierarchical prediction through three sequential but interconnected heads:

- **Anomaly Detection Head:** This binary classification head determines first whether an anomaly is present in the current drilling context. It outputs a probability score indicating the likelihood of anomalous behavior, serving as the primary gate for further analysis.
- **Anomaly Category Head:** Once an anomaly is detected, this classification head identifies the specific type of anomaly from four primary categories: overpull, slack-off, restriction, and stuck

pipe. Each category represents distinct drilling conditions that may lead to stuck pipe incidents, allowing operators to understand the nature of the detected anomaly.

- **Severity Assessment Head:** Following anomaly categorization, this head classifies the severity level of the detected anomaly into three classes: low, medium, and high. This hierarchical assessment enables operators to prioritize their response based on the urgency and potential impact of the detected condition.

Anomaly Classification and Severity Assessment

Anomaly Categories. The system classifies detected anomalies into six primary categories based on their physical manifestations and operational implications:

- **Overpull:** Excessive tension forces during pipe movement, often indicating friction or mechanical resistance.
- **Over-slack:** Insufficient load on the drill string, potentially indicating loss of contact or improper weight transfer.
- **Restriction:** Physical obstructions or tight hole conditions that impede normal pipe movement.
- **Pipe Stalling:** Sudden cessation of pipe rotation or movement due to excessive torque, mechanical binding, or downhole obstructions .
- **Stick and slip:** Intermittent jerky motion characterized by alternating periods of static friction (stick) and sudden movement (slip), typically caused by irregular torque application or borehole conditions.
- **Stuck Pipe:** Complete inability to move or rotate the drill string.

It is important to note that the Overpull, Over-slack, Restriction, Pipe Stalling and Stick and slip anomalies are precursor events that occur before actual Stuck pipe incidents (cf. [Figure 3](#)). Studies show these precursors events can be detected with up to six hours' advance notice^[1], allowing sufficient time for remedial action to prevent stuck pipes.

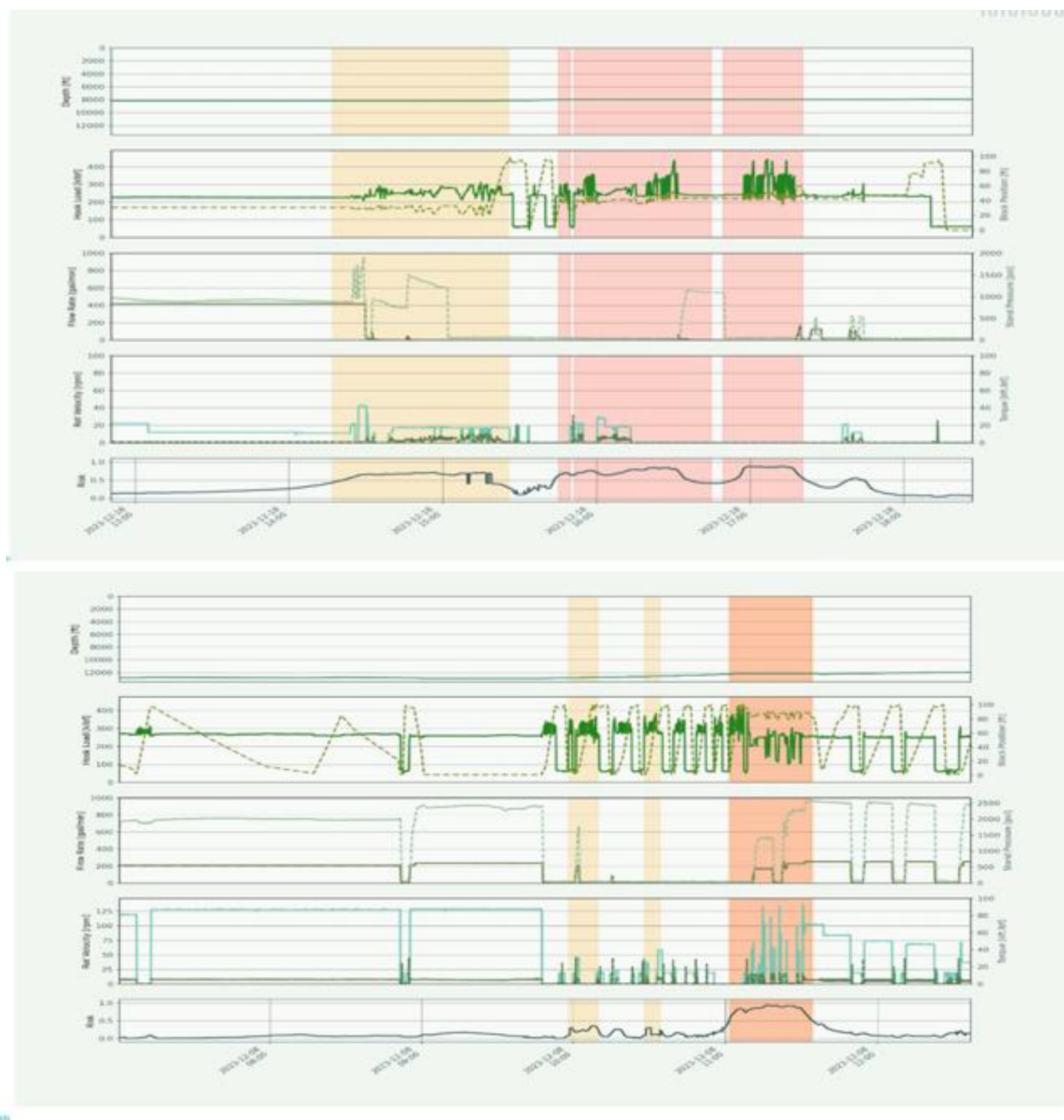


Figure 3—Stuck-pipe detection (in red) preceded by anomalies (in yellow).

Severity Classification. For each detected anomaly, the system assigns a severity level using a three-class classification scheme: low, medium, and high severity. This hierarchical severity assessment enables operators to prioritize their response based on the urgency and potential impact of the detected condition.

The severity classification is based on the magnitude and persistence of the anomalous behavior, with higher severity levels indicating greater deviation from normal operating conditions and increased likelihood of progressing to stuck pipe incidents.

Dynamic Risk Assessment

The dynamic stuck pipe risk indicator integrates outputs from the hierarchical prediction heads to provide a comprehensive assessment of current drilling conditions. The risk assessment follows sequential logic:

anomaly detection triggers category classification, which then determines severity level, and finally contributes to the overall stuck pipe risk calculation.

The risk assessment algorithm considers both the detected anomaly category and its severity level, with higher severity levels within critical categories (such as stuck pipe and restriction) contributing more significantly to the overall risk score. This hierarchical approach ensures that the risk indicator escalates appropriately with increasing anomaly severity and criticality.

Training Methodology

Dataset and Preprocessing. The system was trained and evaluated on a comprehensive dataset comprising 200 drilled sections. The dataset includes diverse drilling scenarios, well profiles, and operational conditions to ensure robust generalization capabilities.

Raw sensor data from drilling operations was collected at high temporal resolution, including parameters such as hook load, torque, standpipe pressure, weight on bit, rotary speed, and flow rates. No preprocessing or feature engineering was applied to the raw data, allowing the deep learning architecture to learn optimal representations directly from the sensor inputs.

Training Configuration. The model was trained using standard deep learning optimization techniques, with careful attention to preventing overfitting given the complexity of the multi-head architecture. Data augmentation techniques specific to time series data were employed to increase the effective size of the training dataset and improve generalization.

The training process incorporated early stopping based on validation performance, and model selection was performed using cross-validation to ensure robust performance estimates across different drilling scenarios.

Real-Time Implementation

The deployment of deep learning systems for real-time drilling operations has gained significant traction in recent industry applications. AI-automated drilling real-time operations centers (RTOC) have successfully integrated machine learning and deep learning approaches for continuous anomaly monitoring, demonstrating the practical feasibility of automated drilling surveillance systems (Benzine et al., 2024a). This work builds upon these developments by providing a comprehensive end-to-end solution specifically optimized for stuck pipe prevention.

The trained DrillNet model was deployed in a real-time monitoring system capable of processing streaming sensor data with minimal latency. The system provides continuous anomaly monitoring during both drilling and tripping operations, with specialized models optimized for each operational phase.

The implementation includes visualization dashboards that present anomaly classifications, severity assessments, and risk indicators in an intuitive format for drilling operators, enabling rapid decision-making based on the system's predictions.

Evaluation

Evaluation Protocol. To evaluate models' performance, we developed the Time Series Segmentation Evaluator (TSSE) (Khaouja et al.(2025); Benzine et al., 2024a), which compares predicted segments with annotated ground truth segments (cf. Figure 4). This segment-based evaluation provides a more accurate assessment of anomaly detection models in continuous data streams than traditional point-by-point methods. As illustrated by the figure below, the TSSE quantifies the overlap (IoU) between predicted and annotated segments, distinguishing between true positives (TP), false positives (FP), and false negatives (FN), and calculates key metrics such as precision, recall, F1-score, and average precision (AP).

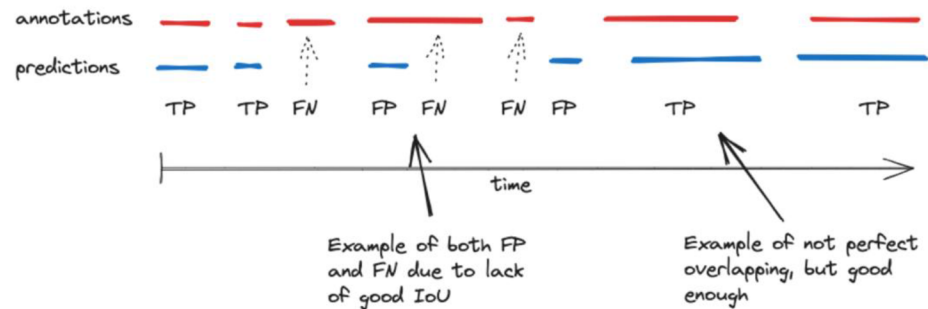


Figure 4—Visual representation of true positives (TP), false positives (FP), and false negatives (FN) in time series segmentation evaluation. The red lines indicate the ground truth segments, while the blue lines indicate the predicted segments. Significant overlap between predictions and annotations results in TPs. In cases where segments do not overlap, FPs, FNs, or both may occur

The TSSE uses an efficient many-to-many matching algorithm, operating in polynomial time, to optimally pair predicted segments with ground truth segments, ensuring accurate detection and classification of events. This approach is particularly effective for detecting sustained anomalies, which often appear as continuous deviations in sensor readings during drilling operations.

Results

Overall Model Performance

The proposed DrillNet system demonstrates significant improvements over existing approaches across all evaluation metrics. Table 3 presents a comprehensive comparison between the legacy model (Benzine et al., 2024a) and our new deep learning system, showing substantial performance gains across precision, recall, and F1-score metrics.

Table 3—Overall Performance Comparison Between Rule Based and the Deep Learning Models for anomaly detection

Model	F1-Score	Precision	Recall
Benzine et al., 2024a	50.1%	53.1%	47.5%
Deep Learning Model	73.0%	74.6%	71.7%

The results show a 23 percentage point improvement in F1-score, from 50% to 73%, representing a 46% relative improvement. The precision increased by 21.5 percentage points, while recall improved by 24.2 percentage points. These improvements demonstrate the effectiveness of the end-to-end deep learning approach in capturing complex patterns in drilling data that were previously undetectable by traditional methods.

Performance Across Different Section Sizes

The system's performance varies across different wellbore section sizes, reflecting the distinct drilling challenges and data characteristics associated with each section. Table 4 provides a detailed breakdown of F1-scores achieved for different hole sizes commonly encountered in drilling operations.

Table 4—Model Performance by Wellbore Section Size

Section Size	F1-Score
24 inch	65.7%
22 inch	83.8%
17.5 inch	78.8%
16 inch	65.7%
12.25 inch	73.0%
8.5 inch	82.5%
6 inch	57.1%

The analysis reveals that the model performs well in intermediate section sizes, with the 22-inch section achieving the highest F1-score of 83.8%, followed closely by the 8.5-inch section at 82.5%. The strong performance in these sections can be attributed to the availability of rich training data and the distinct drilling patterns that characterize these commonly drilled intervals. Conversely, the 6-inch section shows the lowest performance at 57.1%, likely due to the increased complexity and variability in completion-related operations typically associated with smaller diameter sections.

Operational Phase Performance Analysis

The DrillNet system was evaluated across different drilling operational phases to assess its adaptability and effectiveness throughout the entire drilling process. [Table 5](#) summarizes the F1-score performance for each major operational phase.

Table 5—Model Performance by Drilling Operation Type

Operation	F1-Score
Liner	82.2%
Drilling	74.8%
Tripping	72.9%
Casing	70.3%
Completion	66.8%

Liner operations demonstrate the highest prediction accuracy at 82.2%, which can be attributed to the more predictable and structured nature of liner running operations. Drilling operations, representing the core drilling phase, achieve a solid 74.8% F1-score, validating the system's effectiveness in its primary application domain. Tripping operations, involving pipe movement in and out of the wellbore, show good performance at 72.9%, demonstrating the system's capability to detect anomalies during critical pipe handling phases.

Casing operations achieve 70.3% accuracy, while completion operations show the lowest performance at 66.8%. The reduced performance in completion phases reflects the increased complexity and variability of completion operations, which often involve specialized tools and non-standard procedures that may not be well-represented in the training data.

Stuck Pipe Prediction Accuracy

A critical evaluation metric for the system is its ability to accurately predict stuck pipe incidents before they occur. The stuck pipe prediction was evaluated based on whether any anomaly was raised from 6 hours

to 30 minutes before the stuck pipe event occurred. Table 6 compares the stuck pipe prediction accuracy between the old and new systems.

Table 6—Stuck Pipe Prediction Accuracy Comparison

Model	Stuck Pipe Prediction Accuracy
Rule Based Model	50.0%
Deep learning Model	70.0%

The new DrillNet system achieves a 70% accuracy in stuck pipe prediction, representing a 20-percentage point absolute improvement and a 40% relative improvement over the legacy system. This substantial enhancement in predictive capability translates directly to improved operational safety and reduced non-productive time, as operators can now receive more reliable early warnings of impending stuck pipe conditions.

Discussion of Results

The comprehensive evaluation results demonstrate the effectiveness of the proposed end-to-end deep learning approach for stuck pipe prevention. The consistent improvements across all evaluation dimensions—overall performance metrics, section-specific analysis, operational phase performance, and direct stuck pipe prediction—validate the system's robustness and practical applicability.

The variation in performance across different section sizes and operational phases provides valuable insights for system deployment and optimization. The higher performance in liner and drilling operations suggests that these phases benefit most from the deep learning approach, while the lower performance in completion operations indicates opportunities for targeted improvements through specialized training data or model architectures.

The 70% stuck pipe prediction accuracy, while representing a significant improvement over existing methods, also highlights the inherent complexity of the stuck pipe prediction problem. The remaining 30% prediction gap suggests that certain stuck pipe incidents may be caused by factors not fully captured in the current sensor data or may occur with insufficient precursor signals for reliable prediction.

Conclusion

This paper presented a novel end-to-end deep learning system for proactive stuck pipe prevention in drilling operations, addressing critical limitations of existing approaches through direct processing of raw sensor data and hierarchical anomaly detection. The DrillNet architecture achieved significant improvements over traditional methods, with a 73% F1-score representing a 23% relative improvement and 70% stuck pipe prediction accuracy constituting a 20% enhancement over legacy systems.

The multi-head architecture successfully addresses stuck pipe prediction complexity through hierarchical anomaly detection, category classification, and severity assessment, enabling operators to understand not only whether an anomaly exists but also its nature and urgency. Real-world implementation has demonstrated measurable improvements in operational safety and cost reduction, with the system's ability to process raw sensor data eliminating dependency on domain-specific feature engineering.

From a research perspective, this work establishes a new benchmark for stuck pipe prediction and demonstrates that end-to-end learning can effectively capture subtle patterns indicative of stuck pipe precursors. While the 70% prediction accuracy represents substantial progress, approximately 30% of incidents remain unpredictable, suggesting opportunities for future research through integration of additional data sources and advanced architectures.

The successful development and deployment of this system represents a significant advancement in drilling automation technology. By providing early warnings through hierarchical anomaly detection, it offers the potential to transform drilling operations from reactive problem-solving to proactive risk management, establishing a foundation for the next generation of intelligent drilling systems.

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